# 6.1 → Introduction to reserving principles

## 6.1.1 → understanding claims data

→ reported claims = paid claims + case outstanding

→ unpaid claims = case outstanding + IBNR
↳ broad definition
↳ = pure IBNR + IBNER

↓ = ultimate claims − paid claims

⇒ ultimate claims = paid claims + case outstanding + IBNR

↓ = reported claims + IBNR

→ PAID vs Reported
↳ Does not include loss reserves

→ In addition to the estimate itself, when providing an unpaid claims estimate three key dates are required:

- → Accounting date → This date separates paid vs unpaid claim amounts
- → Valuation date → The date through which transactions are included in the estimate. This date can be before, on, or after the accounting date
- → Review date → The cutoff date for information known to the actuary

→ include information in a given AY up to the valuation date

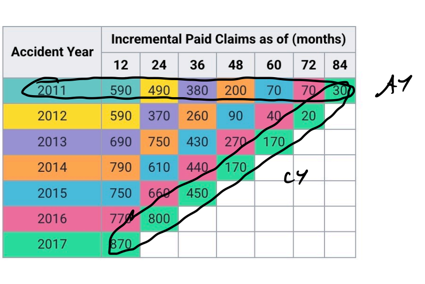
→ Accounting date splits transactions into being paid or unpaid

# 6.2 → Chain ladder method

## 6.2.1 → The development technique

→ Steps:

- →1) Compile claims data
- →2) Calculate age-to-age factors
- →3) Calculate averages
- →4) Select claim development factors
- →5) Select tail factor
- →6) Calculate cumulative development factors
- →7) Project ultimate claims



| Accident Year | Incremental Paid Claims as of (months) | | | | | | |
|---|---|---|---|---|---|---|---|
| | 12 | 24 | 36 | 48 | 60 | 72 | 84 |
| 2011 | 590 | 490 | 380 | 200 | 70 | 70 | 30 |
| 2012 | 590 | 370 | 260 | 90 | 40 | 20 | |
| 2013 | 690 | 750 | 430 | 270 | 170 | | |
| 2014 | 790 | 610 | 440 | 170 | | | |
| 2015 | 750 | 660 | 450 | | | | |
| 2016 | 770 | 800 | | | | | |
| 2017 | 870 | | | | | | |

→ Selecting LDFs key ideas / Goals

- → Smooth progression → Ideally, there is a smooth decrease in factors across development years (i.e. as the claims approach maturity).
- → Stability → w/in each column, a low variance in development factors signifies stability
- → Credibility of experience → Volume + homogeneity determine credibility. If credibility is lacking, then as a last resort industry benchmark development factors may need to be used instead.
- → Changes in patterns → Reviewing age-to-age factors can help identify patterns that reflect internal +/or external changes
- → Applicability of the historical experience → Actuaries should use qualitative information to determine how applicable historical development will reflect future claim development
- → Shock losses / cat losses → Sometimes, we might observe a single very large claim that distorts the development factors for a given year/age.

## 6.2.2 → Analyzing the development method

→ Assumptions + uses

→ The development method calculates claim development factors based on historical data. As such, it makes the following key assumptions:

- →1) Claims recorded to date will continue to develop in a similar fashion in the future
- →2) For accident years that are not fully developed, claims observed thus far give us relevant information on claims that have not yet to be observed
- →3) Throughout the policy period:
  - → the mix of claim types is stable
  - → policy limits + deductibles (if any) are stable
  - → reinsurance retention limits (if any) are stable
  - → There is consistent claims processing (claim settlement rates + case outstanding adequacy).

→ The development method works well:

- →1) when there is a large amount of credible historical claims data available
- →2) For high-frequency, low-severity lines w/ stable + timely reporting of claims does not greatly distort the data
- →3) when the presence or absence of large claims does not greatly distort the data

→ Changes in environment + their impact ☆

→ Increase (decrease) in written exposures impacts ultimate claims in the same direction, but LDFs are unaffected

⇒ assuming exposure dist over the year remains stable
Jan 10→30 Feb 10→30 ... Dec 10→30

→ change in avg accident date ⇒ claims have had less time to develop by year end ⇒ fewer rept claims + less development on known claims
Jan 10→10 Feb 10→15 ... Dec 10→40

⇒ underestimated ultimate losses b/c applying same LDFs to lower values

→ Changes in claims ratios

→ shifts in freq, sev, or LR don't affect LDFs, but ultimate claims estimates are impacted
↳ ⇒ sel still appropriate

change in LR doesn't imply claims behavior is changing
⇒ could be driven by premium, freq, or sev

→ In a Δ, changes in freq + severity show up in the same manner as a change in exposures
⇒ difference is that change in exposures ⇒ change in premium
— — freq/sev ⇏ — — — — —

→ Changes in settlement rates (speed from rept to settled)

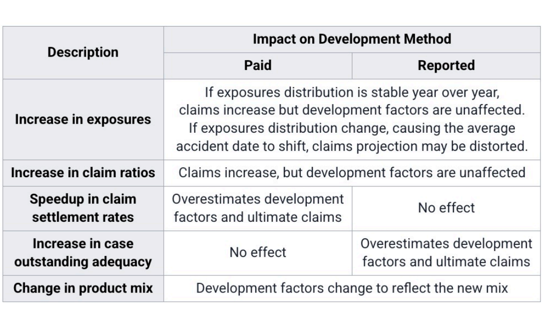
- → impact age-to-age factors + ultimate claims estimates in the same direction as the change in settlement rate w/ the paid dev method
- → no impact in the reported development

→ Changes in case outstanding adequacy

- → No impact on paid dev method
- → Affects reported dev LDFs + ultimate estimates in the same direction of the change

→ Changes in product mix

- → results in bias



| Description | Impact on Development Method | |
|---|---|---|
| | Paid | Reported |
| Increase in exposures | If exposures distribution is stable year over year, claims increase but development factors are unaffected. If exposures distribution change, causing the average accident date to shift, claims projection may be distorted. | |
| Increase in claim ratios | Claims increase, but development factors are unaffected | |
| Speedup in claim settlement rates | Overestimates development factors and ultimate claims | No effect |
| Increase in case outstanding adequacy | No effect | Overestimates development factors and ultimate claims |
| Change in product mix | Development factors change to reflect the new mix | |

# 6.3 → Expected claims method

## 6.3.1 → Expected claims method

→ uses an a priori (initial) estimate + ignores observed claim experience (i.e. only base calculations on full years of data)
⇒ skipping the current, partial year

→1) Using the CY EPs → $\text{ultimate claims} = \underbrace{\frac{\text{Expected Claims}}{\text{Earned Premium}}}_{\text{Expected LR}} \times \text{Earned premium}$

→2) Using CY earned exposures → $\text{ultimate claims} = \underbrace{\frac{\text{Expected claims}}{\text{Earned Exposures}}}_{\text{Expected pure premium}} \times \text{Earned Exposures}$

→ Assumptions + uses

→ The primary assumption of the expected claims technique is that claims paid (or reported) to date for a given AY provide no useful information on future claim payments from that AY. In other words, claims from AY 20XX should be ignored when estimating ultimate losses for AY 20XX.

→ Unlike the development method, the EC method does not assume stability in claims data. Therefore, it is particularly effective in the following scenarios:

- → when an insurer a new LOB or territory
- → when operational or environmental changes make historical data unreliable.
- → when data is limited or unreliable
- → when development factors for early maturities are highly leveraged

→ The EC method provides a stable estimate for ultimate claims b/c it is not influenced by the variability in early claims data. However, its stability also means it may not respond quickly to emerging claims experience, potentially resulting in a lag in recognizing significant changes in actual experience.

→ Also, the expected claims method relies on the actuary's ability to determine a reasonable expected claims ratio. Therefore, if a reasonable claims ratio cannot be identified, this method may not perform effectively.

→ Determining the ECR

→ Actuaries can determine the ECR in various ways. Some methods involve complex statistical modeling, while others are more straightforward. In general, determining an appropriate ECR involves multiple steps + adjustments to account for historical experience, trends, + other external factors:

- →1) Adjust the historical data → This includes on-leveling the premium, adjusting for legal + environmental changes, developing historical losses, + applying trends
- →2) Calculate the claims ratio → Once the data is adjusted, the claims ratio for each historical year are calculated as the ultimate claims divided by the exposure base
- →3) Calculate average claims ratios → The actuary takes different averages (e.g. straight average, medial average, etc.) of the adjusted claims ratios
- →4) Select the ECR → The final step is to choose an ECR based on the calculated averages + actuarial judgement

→ Two rules for trending an ECR

- →1) The claims must be trended to the AY for which you are projecting ultimate claims
- ? →2) The premiums can be trended to any one date, as long as the EP you multiply the ECR by is trended to that same date

→ Expected claims method

- → If finding ECR, don't use data from the year in which we are estimating ultimate claims
- → once have ECR (or if it's given), just use regular CY/AY EP (i.e. not trended or on-leveled)

## 6.3.2 → Analyzing the expected claims method

→ Main idea → Even though use historical data when determining the ECR, the actuary makes a final judgemental selection
⇒ stable is advantage, unresponsive is disadvantage ]

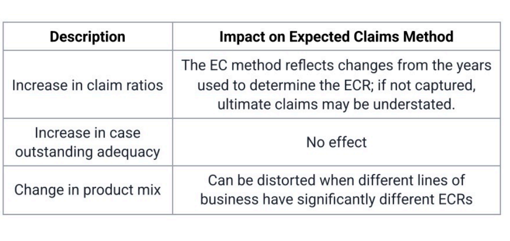
→ Changes in environment + their impact ☆

→ Changes in claims ratios
→ not responsive to changes in freq, sev, or LR that occur during the estimation year
⇒ "out of sync" w/ the actual claims experience

→ Changes in case outstanding adequacy
→ Not affected ⇒ advantage of this method
↳ doesn't affect ultimate estimates, only how conservative reserves are set

→ Changes in product mix
→ Does affect the accuracy of the ECR, especially w/ when LOBs have different ECRs
→ To remedy this, can do a weighted Average ECR (weighted by EP)

→ Summary → implicitly ignores all environmental changes
→ only exception is shifting claims ratios in recent years, which the ECR may not pick up on



| Description | Impact on Expected Claims Method |
|---|---|
| Increase in claim ratios | The EC method reflects changes from the years used to determine the ECR; if not captured, ultimate claims may be understated. |
| Increase in case outstanding adequacy | No effect |
| Change in product mix | Can be distorted when different lines of business have significantly different ECRs |

# 6.4 → Bornhuetter-Ferguson + Benktander methods

## 6.4.1 → Bornhuetter-Ferguson Method

→ Overview → The BF method is a blend of the development method + the EC method
→ Splits ultimate claims estimate into actual reported (paid) claims + expected reported (unpaid) claims, giving more weight to actual claims over time

→ Formulas → $\text{ultimate claims} = \text{Actual reported claims} + \underbrace{\text{Expected claims} \times \%\,\text{unreported}}_{\text{Expected unreported claims}}$

$\text{ultimate claims} = \text{Actual paid claims} + \underbrace{\text{Expected claims} \times \%\,\text{Unpaid}}_{\text{Expected unpaid claims}}$

where

$$\%\,\text{unreported} = 1 - \frac{1}{\text{Reported Age-to-Ult Factor}}$$

$$\%\,\text{unpaid} = 1 - \frac{1}{\text{Paid Age-to-Ult Factor}}$$

→ credibility weighted average

$$\text{ultimate claims} = \text{Development ultimate claims}\,\underset{\%\text{ reported}}{\overset{Z}{\left(\frac{1}{CDF}\right)}} + \text{Expected claims ultimate claims}\,\underset{\%\text{ unreported}}{\overset{1-Z}{\left(1-\frac{1}{CDF}\right)}}$$

→ After projecting the ultimate claims, we subtract the actual reported claims to estimate IBNR, or subtract the actual paid claims to estimate the unpaid claims:

IBNR Estimate = Ultimate claims − Actual reported claims
↓ = Expected claims × % unreported

Unpaid Claims Estimate = Ultimate claims − Actual paid claims
↓ = Expected claims × % unpaid

→ Steps → To determine the reserve using the BF technique:

- →1) Estimate ultimate losses using the expected claims method
- →2) Calculate the cumulative loss development factors (CDF) using the development method
- →3) Estimate ultimate losses using one of the top two formulas above
- →4) To estimate IBNR, subtract actual reported losses; to estimate unpaid claims, subtract actual paid claims

→ Assumptions + uses

→ As a hybrid between the EC + development methods, the BF method inherits some assumptions from each of them:

☆ { →1) Like the EC method, the BF method assumes that the unreported (or unpaid) claims will develop based on the expected claims, which means the claims reported (or paid) to date provide no information for further development.
→2) Then, it assumes that future claims, both reported + paid, develop in the same pattern as in the development method

→ B-F method final result → $\underset{= 1/CDF}{\text{Dev method ult claims}\ (Z)} + \text{EC method ult claims} \times \underset{= 1 - \frac{1}{CDF}}{(1 - Z)}$
↳ ECR × Selected ult claims ↳ from dev method

⇒ Same for ALAE too! except selected ALAE ratio

If solving for this, skip year of interest

## 6.4.2 → Benktander Method

→ Main advantage of B-F method → stability ⇒ places more emphasis on the expected claims in the early development years + less on the emerging actual claims
⇒ early on, more heavily influenced by the actuarial judgement of the EC method, which may not always be beneficial

→ The Benktander method combines the BF technique + the development technique in a credibility-weighted average. Specifically, while the BF technique depends on the EC method in estimating the expected claims, the Benktander method utilizes the projected ultimate claims from an initial BF projection:

$$\text{Benktander ultimate claims} = \text{Development ultimate claims} \times \left(\frac{1}{CDF}\right) + \text{BF ultimate claims}\left(1 - \frac{1}{CDF}\right)$$

→ Thus, the Benktander method is essentially an iterative version of the BF technique, w/ the main difference being the derivation of expected claims. After sufficient iterations, the Benktander projection of ultimate claims will converge to the projected ultimate claims generated by the development technique.

→ Note that, compared to the BF method, the Benktander method places more weight on the development technique, making it more responsive to the changes in the claims data. Yet, it still provides more stable estimates than the development technique.

## 6.4.3 → Analyzing the BF + Benktander methods

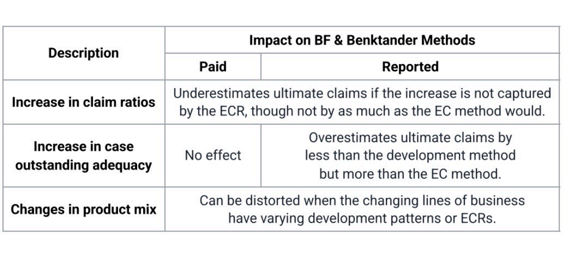
→ Main idea → middle ground between past expectations + present data
→ less volatile than the development method, but also potentially less accurate in rapidly evolving environments if not carefully updated

→ Changes in environment + their impact ☆

→ Changes in claims ratios
→ Both methods are sensitive to ECR changes
→ If claims ratios increase (due to inflation or a large claim), but the ECR is not updated, the expected claims are understated
⇒ underestimate ultimate claims
→ — — — decrease — — — — — — overstated
⇒ overestimate — — —

→ Changes in case outstanding adequacy
→ only affect reported versions (i.e. paid versions are unaffected)
→ If case reserves become larger (more adequate) ⇒ reported claims increase ⇒ dev factors increase ⇒ potential overestimation of ultimate claims
→ — — — smaller (less — —) ⇒ — — — decrease ⇒ — — — decrease ⇒ — — — underestimation — — —
→ Since both methods combine development-based estimates w/ expected claims, they provide some buffering against these distortions
⇒ concern is if changes are sudden or significant

→ Changes in product mix
→ shifts can introduce bias, two main concerns:
→ varying development factors: e.g. if shift to writing more long-tailed after only writing short-tailed lines, the existing development assumptions may no longer be appropriate, which can affect the development-based portion of the estimates
→ varying claims ratios: If the new LOBs have materially different ECRs + the expected claims calculation is not updated or properly segmented, then the expected claims portion may be inaccurate, leading to biased estimates
→ both methods assume a relatively stable + homogeneous mix of business ⇒ new segments w/ significantly different behavior can lead to breakdown of the assumptions



| Description | Impact on BF & Benktander Methods | |
|---|---|---|
| | Paid | Reported |
| Increase in claim ratios | Underestimates ultimate claims if the increase is not captured by the ECR, though not by as much as the EC method would. | |
| Increase in case outstanding adequacy | No effect | Overestimates ultimate claims by less than the development method but more than the EC method. |
| Changes in product mix | Can be distorted when the changing lines of business have varying development patterns or ECRs. | |

# 6.5 → Cape Cod method

## 6.5.1 → Cape Cod method

→ Overview → Similar to BF method, except the ECR is determined

→ Ultimate claims = Actual Reported Claims + Expected unreported claims

→ formula → $\text{Expected unreported claims} = \left(\text{On-level EP} \times \text{Cape Cod ECR}\right) \times \%\,\text{unreported}$

→ The Cape Cod ECR is determined as follows:

$$\text{Cape Cod ECR} = \frac{\text{Total reported claims to date}}{\text{Total used-up premium}}$$

$$\downarrow \quad = \frac{\sum \text{reported claims}}{\sum\left(\text{on-level EP} \times \%\,\text{reported}\right)}$$

↳ always determined from data ⇒ places less emphasis on actuarial judgement
+ data from the year in which the estimate is made is included

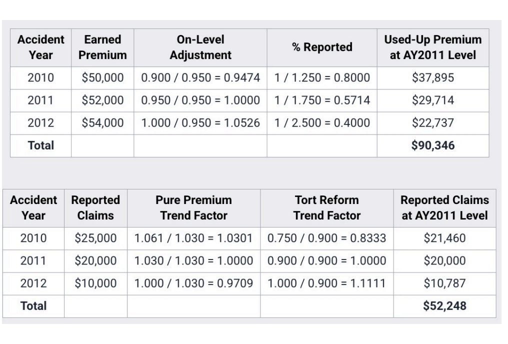
→ Assumptions + uses

→ The Cape Cod method operates on the assumption that unreported claims will develop based on expected claims, which are determined using reported claims + EP. It is commonly used by reinsurers + is applicable to all lines of insurance, including both short-tail + long-tail lines.

→ An advantage of the Cape Cod method, when compared to the development technique, is its resilience against early random fluctuations in the development of an AY or time interval. This is similar to the EC technique.

→ Adjusting data

→ example →



| Accident Year | Earned Premium | On-Level Adjustment | % Reported | Used-Up Premium at AY2011 Level |
|---|---|---|---|---|
| 2010 | $50,000 | 0.900 / 0.950 = 0.9474 | 1 / 1.250 = 0.8000 | $37,895 |
| 2011 | $52,000 | 0.950 / 0.950 = 1.0000 | 1 / 1.750 = 0.5714 | $29,714 |
| 2012 | $54,000 | 1.000 / 0.950 = 1.0526 | 1 / 2.500 = 0.4000 | $22,737 |
| Total | | | | $90,346 |

| Accident Year | Reported Claims | Pure Premium Trend Factor | Tort Reform Trend Factor | Reported Claims at AY2011 Level |
|---|---|---|---|---|
| 2010 | $25,000 | 1.061 / 1.030 = 1.0301 | 0.750 / 0.900 = 0.8333 | $21,460 |
| 2011 | $20,000 | 1.030 / 1.030 = 1.0000 | 0.900 / 0.900 = 1.0000 | $20,000 |
| 2012 | $10,000 | 1.000 / 1.030 = 0.9709 | 1.000 / 0.900 = 1.1111 | $10,787 |
| Total | | | | $52,248 |

→ OR → we could also adjust the premiums + claims to AY 2012 levels, determine the ECR, + then adjust the ECR to AY 2011 levels before multiplying it to the AY 2011 EP.

$$\text{ECR at 2011 level} = \frac{\text{Reported claims at 2011 level}}{\text{used-up premiums at 2011 level}}$$

$$= \frac{\dfrac{\text{Reported claims at 2012 level}}{\text{PP trend factor}_{2011} \times \text{Tort reform factor}_{2011}}}{\dfrac{\text{used-up premiums at 2012 levels}}{\text{On-level adjustment}_{2011}}}$$

"relative to 2011"

$$= \frac{\text{Reported claims at 2012 level}}{\text{used-up premiums at 2012 level}} \times OLF_{2011} \times \frac{1}{\text{PP Trend factor}_{2011}} \times \frac{1}{\text{Tort reform factor}_{2011}}$$

→ Cape Cod method → ① OLEP ② cumulative reported losses ③ LDF ④ % reported 1/(3) ⑤ used-up premium (1) × (4) } ⇒ Everything for premium, no development for losses

(be sure to check "as of date" * NOT developed) only use this for (4)

↓ Trend to level of interest

(6) Cape Cod ECR = Σ(2) / Σ(5) (not a selection, data-driven)

☆ ⇒ have to then adjust to be at level of interest in terms of TREND (i.e. Premium trend / loss trend)
⇒ w/ my strat, we only adjust if going "backwards" → need to do the opposite of when trending "forward"

⇒ Estimated unreported claims = ① × (6) × (1 − (4))

→ Cape-Cod method → Brings losses + premiums to current + consistent basis before finding Cape Cod ECR ⇒ TREND + reforms first
↳ (just to most recent year)
⇒ Trending here is separate from trending for the indication / new rates

## 6.5.2 → Analyzing the Cape Cod method

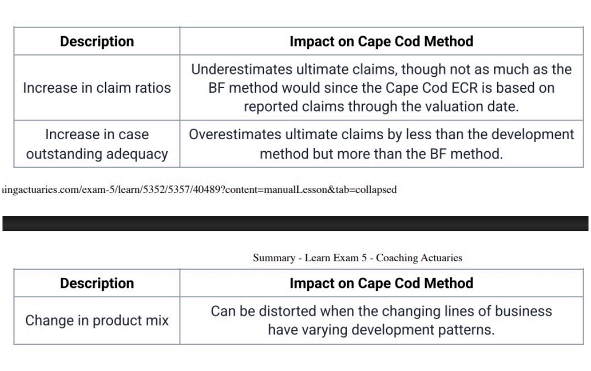
→ similar impacts to the BF method

→ Changes in environment + their impact ☆

→ Changes in claims ratios
→ Derives its ECR from actual reported data, giving considerable more weight to recent AYs, which makes it more reactive to changes in claims ratios than the BF-method
→ However it lacks the full adaptability of the development technique, which directly reflects observed loss emergence
⇒ if claims ratios are increasing, the Cape Cod method may not keep up fast enough ⇒ underestimation of ultimate claims

→ Changes in case outstanding adequacy
→ This method relies on reported claims data ⇒ changes in case reserve adequacy directly impacts the model's output
→ e.g.) strengthen case reserves ⇒ reported claims appear larger ⇒ overestimate ultimate losses
→ less reliable than BF method when case adequacy is shifting, but less vulnerable to these distortions than the reported dev method

→ Changes in product mix
→ Assumes that past experience provides a reliable basis for estimating future claims
⇒ may produce estimates based on older, less representative experience if the book is changing



| Description | Impact on Cape Cod Method |
|---|---|
| Increase in claim ratios | Underestimates ultimate claims, though not as much as the BF method would since the Cape Cod ECR is based on reported claims through the valuation date. |
| Increase in case outstanding adequacy | Overestimates ultimate claims by less than the development method but more than the BF method. |

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| Description | Impact on Cape Cod Method |
|---|---|
| Change in product mix | Can be distorted when the changing lines of business have varying development patterns. |